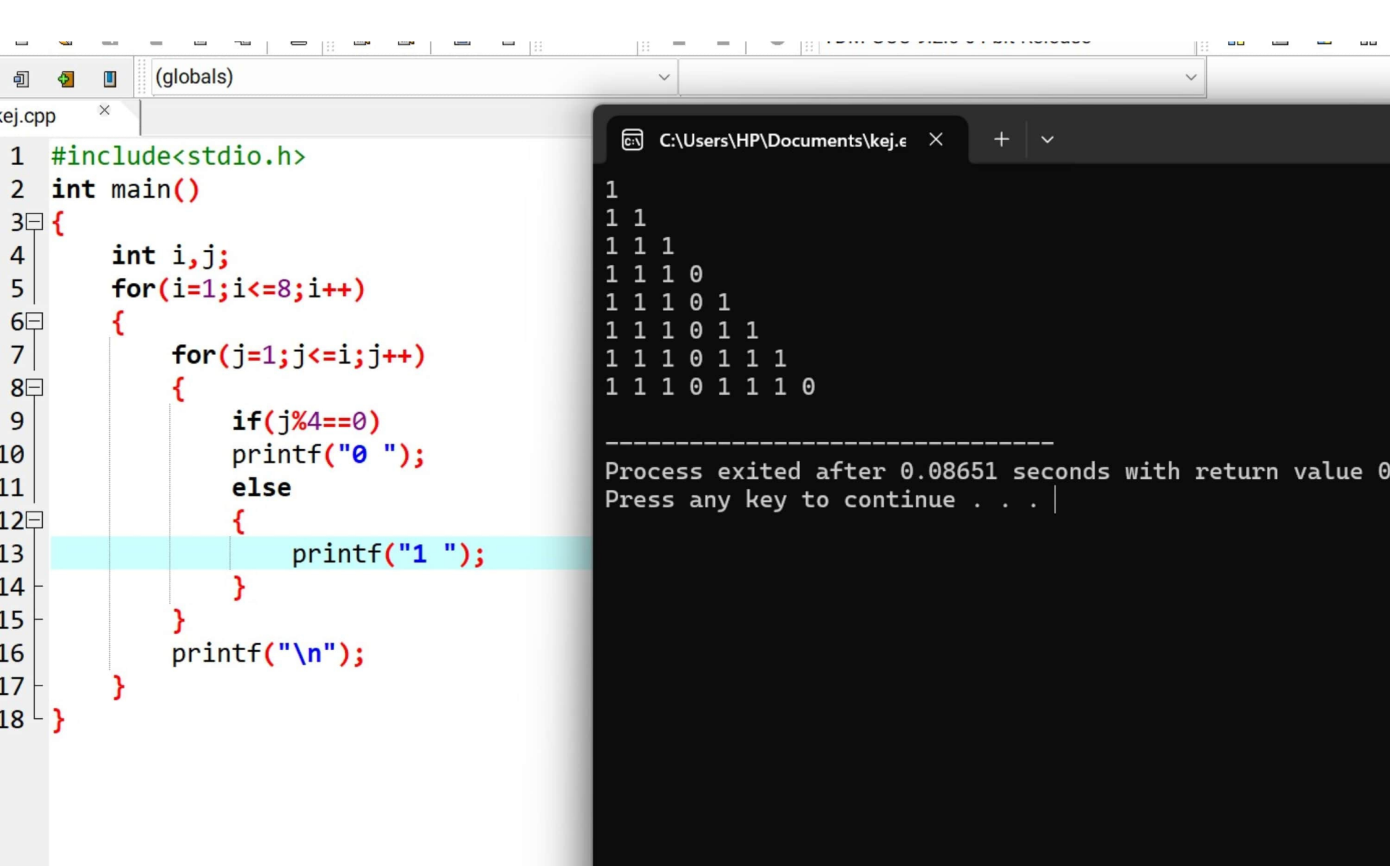

(globals)
kej.cpp
#include<stdio.h>
int main()
{
int i,j;
for(i=1;i<=8;i++)
{
for(j=1;j<=i;j++)
{
if(j%4==0)
printf("0 ");
else
{
printf("1 ");
}
}
printf("\n");
}
}
C:\Users\HP\Documents\kej.e
1
1 1
1 1 1
1 1 1 0
1 1 1 0 1
1 1 1 0 1 1
1 1 1 0 1 1 1
1 1 1 0 1 1 1 0
Process exited after 0.08651 seconds with return value 0
Press any key to continue . . .

CamScanner

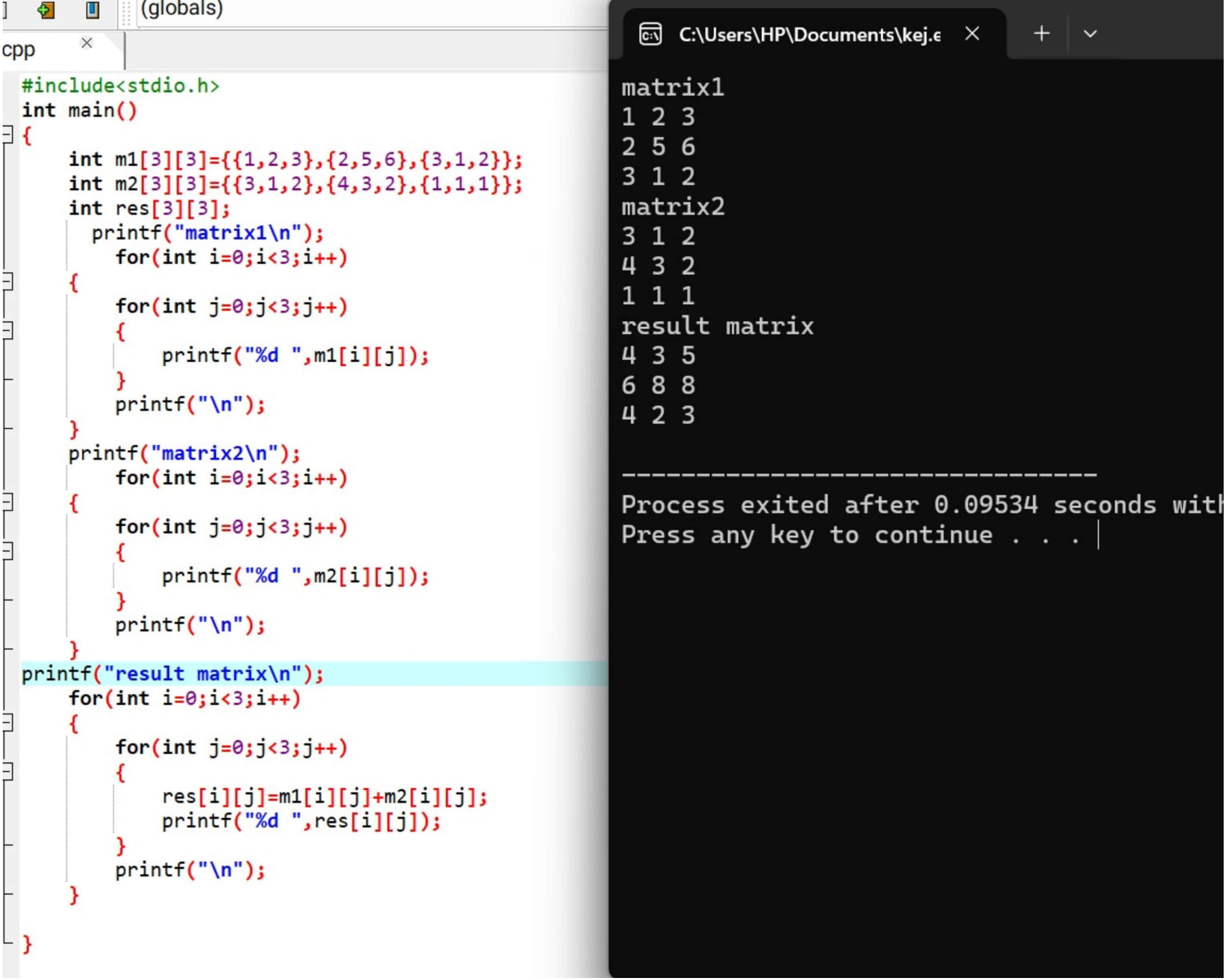
ej.cpp

```
1  #include<stdio.h>
2  int main()
3  {
4      int i,j;
5      for(i=1;i<=5;i++)
6      {
7          for(j=1;j<=i;j++)
8          {
9              if(j%2==0)
10             {
11                 printf("0 ");
12             }
13             else
14             {
15                 printf("1 ");
16             }
17         }
18         printf("\n");
19     }
20 }
```

```
1
1 0
1 0 1
1 0 1 0
1 0 1 0 1

--------------------------------
Process exited after 0.08674 seconds with return value
Press any key to continue . . .
```

CamScanner


(globals)
cpp
#include<stdio.h>
int main()
{
int m1[3][3]={{1,2,3},{2,5,6},{3,1,2}};
int m2[3][3]={{3,1,2},{4,3,2},{1,1,1}};
int res[3][3];
printf("matrix1\n");
for(int i=0;i<3;i++)
{
for(int j=0;j<3;j++)
{
printf("%d ",m1[i][j]);
}
printf("\n");
}
printf("matrix2\n");
for(int i=0;i<3;i++)
{
for(int j=0;j<3;j++)
{
printf("%d ",m2[i][j]);
}
printf("\n");
}
printf("result matrix\n");
for(int i=0;i<3;i++)
{
for(int j=0;j<3;j++)
{
res[i][j]=m1[i][j]+m2[i][j];
printf("%d ",res[i][j]);
}
printf("\n");
}
}
C:\Users\HP\Documents\kej.e
matrix1
1 2 3
2 5 6
3 1 2
matrix2
3 1 2
4 3 2
1 1 1
result matrix
4 3 5
6 8 8
4 2 3
Process exited after 0.09534 seconds with
Press any key to continue . . .

CamScanner

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TDM-GCC 9.2.0 64-bit Release

(globals)

[*] kej.cpp

```
#include<stdio.h>
int main()
{
    int marks[4][2];
    int total;
    int sumMid=0,sumFinal=0;
    int highMid,highFinal,highAll;
    int lowMid,lowFinal,lowAll;
    for(int i=0;i<4;i++)
    {
        for(int j=0;j<2;j++)
        {
            if(j==0)
            printf("enter mid marks for student %d ",i+1);
            else
                printf("enter final marks for student %d ",i+1);
            scanf("%d",&marks[i][j]);
        }
    }
    printf("total marks of each student\n");
    for(int i=0;i<4;i++)
    {
        total=marks[i][0] +marks[i][1];
        printf("student %d=%d\n",i+1,total);
    }
    highMid = marks[0][0];
    lowMid = marks[0][0];
    highFinal = marks[0][1];
    lowFinal = marks[0][1];
    for (int i = 0; i < 4; i++)
    {
        if (marks[i][0] > highMid)
            highMid = marks[i][0];
        if (marks[i][0] < lowMid)
            lowMid = marks[i][0];
        if (marks[i][1] > highFinal)
            highFinal = marks[i][1];
        if (marks[i][1] < lowFinal)
            lowFinal = marks[i][1];
```

Search

11:08 AM
10/18/2026

CamScanner

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TDM-GCC 9.2.0 64-bit Release

(globals)

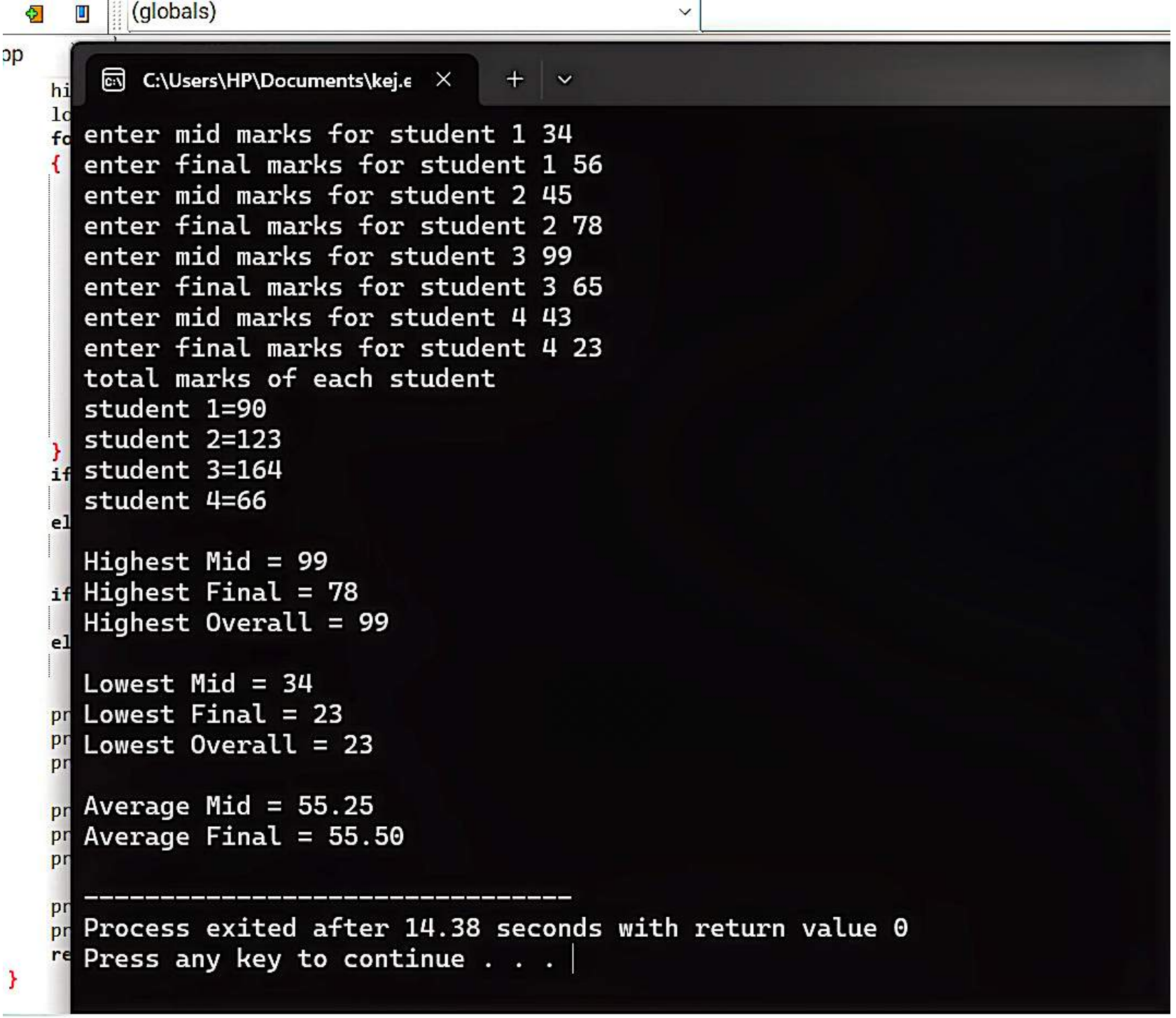
[*] kej.cpp

```
28      highFinal = marks[0][1];
29      lowFinal = marks[0][1];
30      for (int i = 0; i < 4; i++)
31      {
32          if (marks[i][0] > highMid)
33              highMid = marks[i][0];
34          if (marks[i][0] < lowMid)
35              lowMid = marks[i][0];
36          if (marks[i][1] > highFinal)
37              highFinal = marks[i][1];
38          if (marks[i][1] < lowFinal)
39              lowFinal = marks[i][1];
40
41          sumMid = sumMid + marks[i][0];
42          sumFinal = sumFinal + marks[i][1];
43      }
44      if (highMid > highFinal)
45          highAll = highMid;
46      else
47          highAll = highFinal;
48
49      if (lowMid < lowFinal)
50          lowAll = lowMid;
51      else
52          lowAll = lowFinal;
53
54      printf("\nHighest Mid = %d\n", highMid);
55      printf("Highest Final = %d\n", highFinal);
56      printf("Highest Overall = %d\n", highAll);
57
58      printf("\nLowest Mid = %d\n", lowMid);
59      printf("Lowest Final = %d\n", lowFinal);
60      printf("Lowest Overall = %d\n", lowAll);
61
62      printf("\nAverage Mid = %.2f\n", sumMid / 4.0);
63      printf("Average Final = %.2f\n", sumFinal / 4.0);
64      return 0;
65  }
66
```

Search

11:09 AM
10/18/2026

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(globals)
C:\Users\HP\Documents\kej.e
enter mid marks for student 1 34
enter final marks for student 1 56
enter mid marks for student 2 45
enter final marks for student 2 78
enter mid marks for student 3 99
enter final marks for student 3 65
enter mid marks for student 4 43
enter final marks for student 4 23
total marks of each student
student 1=90
student 2=123
student 3=164
student 4=66
Highest Mid = 99
Highest Final = 78
Highest Overall = 99
Lowest Mid = 34
Lowest Final = 23
Lowest Overall = 23
Average Mid = 55.25
Average Final = 55.50
Process exited after 14.38 seconds with return value 0
Press any key to continue . . .

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Q5: 3D Array

this array has 2-semesters, 3-students and 3-subjects

total no: of elements = 2x3x3 = 18

two layers

Layer 0 - Semester 1 :

student	subject 0	subject 1	subject 2
student 0	85	91	72
student 1	73	84	93
student 2	96	81	87

Layer 1 - Semester 2 :

	subject 0	subject 1	subject 2
student 0	88	93	81
student 1	90	85	79
student 2	82	98	90

a) index & value of highest mark

the highest mark is 98 found in semester 2, student 2, subject 1.

index: marks [1] [2] [1]

value: 97

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b) marks of student 1 in semester 2 across all subjects.

marks [1] [0] [0] = 88

marks [1] [0] [1] = 95

marks [1] [0] [2] = 81

c) marks of each student in semester 1.

* student 1 : 85 + 91 + 72 = 248
* student 2 : 73 + 84 + 93 = 250
* student 3 : 96 + 81 + 87 = 264

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Q6: 3D Array

this array contains 2-layers, 3 rows and 3 columns

total elements: $2 \times 3 \times 3 = 18$

Layer 0

arr[i][j][k]	column 0	column 1	column 2
Row 0	arr[0][0][0] = 7	arr[0][0][1] = 15	arr[0][0][2] = 12
Row 1	arr[0][1][0] = 35	arr[0][1][1] = 30	arr[0][1][2] = 28
Row 2	arr[0][2][0] = 65	arr[0][2][1] = 72	arr[0][2][2] = 82

Layer 1

arr[i][j][k]	column 0	column 1	column 2
Row 0	arr[1][0][0] = 23	arr[1][0][1] = 75	arr[1][0][2] = 12
Row 1	arr[1][1][0] = 52	arr[1][1][1] = 29	arr[1][1][2] = 9
Row 2	arr[1][2][0] = 61	arr[1][2][1] = 15	arr[1][2][2] = 7

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Q7: 3D Array

this array contains 3-layers, 2rows and 4 columns

total elements : 3x2x4 = 24

Layer 0

	column 0	column 1	column 2	column 3
Row 0	arr[0][0][0] = 5	arr[0][0][1] = 3	arr[0][0][2] = 12	arr[0][0][3] = 8
Row 1	arr[0][1][0] = 7	arr[0][1][1] = 4	arr[0][1][2] = 9	arr[0][1][3] = 13

First element : arr[0][0][0] = 5

Last element : arr[0][1][3] = 15

Layer 1

	column 0	column 1	column 2	column 3
Row 0	arr[1][0][0] = 9	arr[1][0][1] = 18	arr[1][0][2] = 23	arr[1][0][3] = 39
Row 1	arr[1][1][0] = 15	arr[1][1][1] = 21	arr[1][1][2] = 22	arr[1][1][3] = 32

First element : arr[0][0][0] = 9

Last element : arr[1][1][3] = 32

Layer 2

	column 0	column 1	column 2	column 3
Row 0	arr[2][0][0] = 7	arr[2][0][1] = 2	arr[2][0][2] = 15	arr[2][0][3] = 29
Row 1	arr[2][1][0] = 5	arr[2][1][1] = 12	arr[2][1][2] = 25	arr[2][1][3] = 31

First element : arr[0][0][0] = 7

Last element : arr[2][1][3] = 31

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